SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I _____**G.Yashmita (192511032)**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.

5. **Cloud Security Fundamentals: Protecting Data and Applications**
    - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
    - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
    - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
    - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
    - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **Server Virtualization and Its Role in Cloud Computing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. **Cloud Security: Threats and Protection Mechanisms**

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. Software as a Service (SaaS): Architecture and Business Applications

Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Technical Presentation

Server Virtualization and Its Role in Cloud computing

Introduction

* Traditional data centers often dedicate physical servers to individual workloads.
* Dedicated infrastructure can leave CPU and memory underutilized.
* Key components: Physical host, hypervisor, virtual machines and cloud computing management.
* Virtualization also simplifies workload portability across hosts.

Aim & objectives

Aim: To explain how Server virtualization enables the scalable, shared and efficient infrastructure that and efficient infrastructure that underpins cloud computing

* Describe virtualization architecture & its layers
* Differentiate Type 1 & Type 2 hypervisors.
* Explain virtual machines & resource allocation.
* Understand how virtualization supports.

Core Technical Explanation

Physical server -> Provides CPU, memory, storage and network resources.

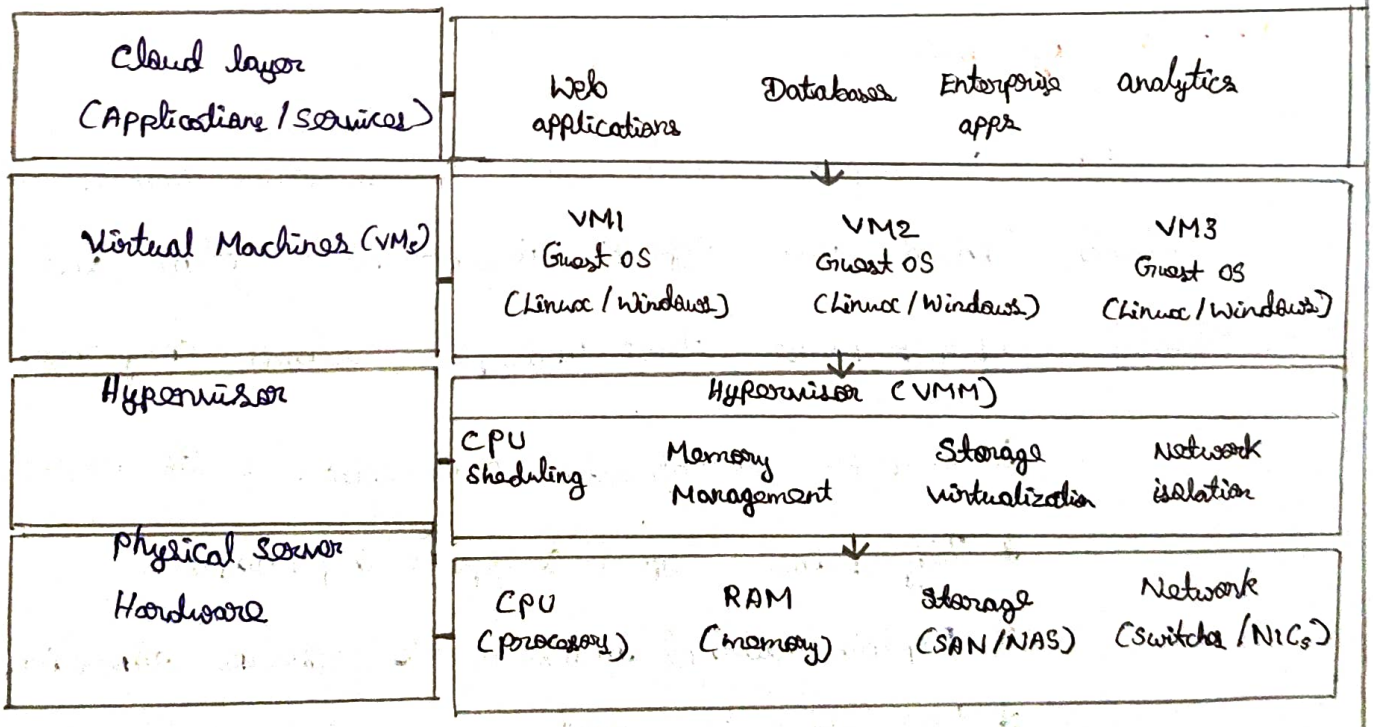
Hypervisor -> Creates and manages multiple virtual machines on a physical server

Virtual machines (VMs) -> Run independent operating systems and applications in isolated environments.

Resource allocation -> Dynamically allocates CPU, memory, storage and network resources to VMs.

VM Isolation -> keeps workloads separated to improve security, stability & reliability.

Virtualization architecture



Real-world use case in Cloud computing

Amazon EC2 (AWS)

* Each EC2 instance is a VM on a shared Physical host.

* Instance types set VCPU, memory & network capacity.

* Pay-per-use billing and fast auto-scaling.

* Azure (Hyper-V) and Google compute engine (KVM) follow the same model.

* Elastic provisioning matches compute capacity to changing demand.

Benefits in Data centers

- Higher utilization
  - Server consolidation
- Lower cost
  - Less hardware, power, cooling
- Scalability
  - Provision VMs in minutes
- Isolation and resilliance
  - Live migration, fast recovery

Conclusion and key Takeaways

* Virtualization decouples software from hardware, making cloud elasticity and multitenancy possible.

* The Hypervisor is the core component : it creates, VMs, allocates

* Virtualization is the foundation of IaaS & Public cloud Platforms.

* Efficient virtualization improves both infrastructure agility and utilization

Topic :

15 Server Virtualization and Its Role in Cloud Computing

Introduction

Server virtualization is a technology that allows multiple virtual machines (VMs) to operate on a single physical server. It uses a software layer called a hypervisor to divide and manage hardware resources such as CPU, memory, storage, and networking. Each virtual machine functions like an independent computer and can run its own operating system and applications.

Traditionally, organizations used separate physical servers for different applications, which increased hardware costs and power consumption. Server virtualization solves this problem by allowing multiple workloads to share the same physical infrastructure. It improves resource utilization, simplifies server management, and reduces operational expenses. Today, virtualization plays a significant role in cloud computing, data centers, enterprise applications, and software development environments.

Objective

- To understand the concept and importance of server virtualization in modern computing.
- To study virtual machines and their role in creating independent computing environments.
- To explain the functions and types of hypervisors used in virtualization.
- To understand the architecture and working process of a virtualized server.
- To examine how CPU, memory, storage, and networking resources are allocated among virtual machines.
- To analyze the advantages of virtualization, including cost reduction, scalability, flexibility, and improved resource utilization.
- To understand how server virtualization supports cloud computing services and data center operations.
- To identify practical applications of virtualization in educational institutions, businesses, and cloud service providers.

Core Technical Explanation

Server virtualization is the process of creating multiple virtual servers from the resources of a single physical server. Each virtual server is called a virtual machine and operates independently within the available hardware capacity. A virtual machine consists of virtual CPU, memory, storage, network interfaces, an operating system, and applications.

The hypervisor is the central component of server virtualization. It acts as an interface between the physical hardware and virtual machines. It creates and manages VMs, distributes computing resources, monitors their performance, and helps maintain isolation between different workloads. Hypervisors are generally classified into two types: Type 1 hypervisors, which run directly on physical hardware, and Type 2 hypervisors, which run on top of a host operating system.

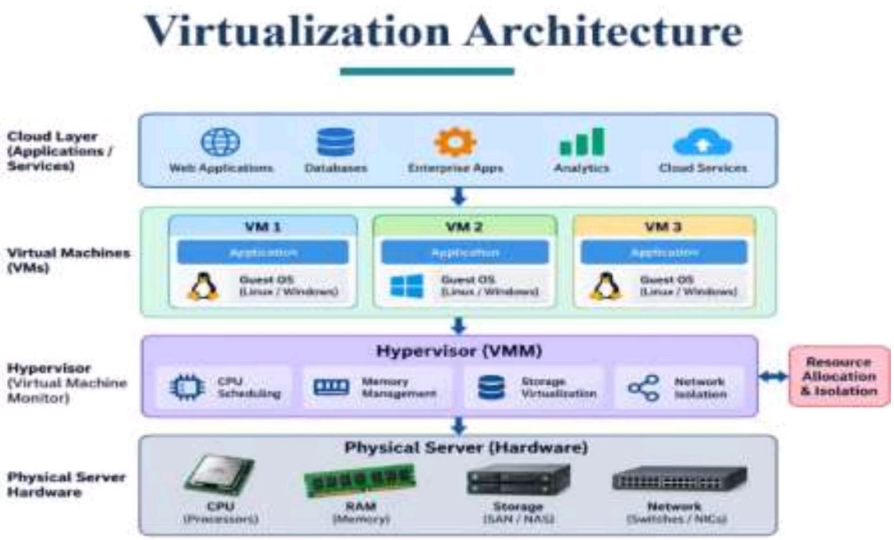
Virtualization improves hardware utilization by allowing multiple applications to share the same physical server instead of requiring separate machines. For example, a company can run its website, database, application server, and testing environment on different VMs hosted by one powerful physical server. The hypervisor allocates resources according to workload requirements and available capacity.

In cloud computing, virtualization enables service providers to create and manage computing environments quickly. It supports resource pooling, isolation, scalability, and rapid provisioning of virtual servers. However, effective management is necessary to prevent resource contention, maintain security, and reduce the impact of physical server failures.

Architecture or Process Diagram

The architecture of server virtualization consists of physical hardware, a hypervisor, virtual machines, guest operating systems, and applications.

Physical Hardware → Hypervisor → Virtual Machines → Guest Operating Systems → Applications



Real-World Use Case

Server virtualization is widely used in cloud data centers operated by providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud. These providers use virtualized infrastructure and other resource-management technologies to deliver computing services to customers on demand.

For example, consider a college that requires separate environments for its website, student management system, online examination portal, and database. Traditionally, the college might use four physical servers, resulting in higher hardware costs, electricity consumption, cooling requirements, and maintenance efforts.

Using server virtualization, the college can host these workloads in separate virtual machines on a suitably configured physical server. Each VM can run its own operating system and applications, while the hypervisor manages resource allocation and helps isolate the workloads. If the demand for an application increases, administrators can adjust its allocated resources or deploy additional infrastructure when necessary.

This approach reduces hardware requirements, improves resource utilization, and simplifies administration. In large cloud environments, similar principles enable providers to host many customer workloads across extensive physical infrastructure while offering flexible computing resources.

Conclusion and Key Takeaway

Server virtualization is a fundamental technology that improves the efficiency and flexibility of modern computing infrastructure. It enables multiple virtual machines to run on a single physical server while sharing CPU, memory, storage, and networking resources. The hypervisor manages these resources and provides the foundation for creating and operating independent virtual environments.

The major benefits of server virtualization include better hardware utilization, reduced infrastructure costs, faster deployment, simplified management, improved flexibility, and easier backup and recovery. It also supports important cloud computing capabilities such as resource pooling, rapid provisioning, and scalable infrastructure.

Despite these benefits, organizations must manage security risks, resource contention, performance overhead, and physical host failures. Proper monitoring, access controls, backups, and high-availability mechanisms help address these challenges.

Key Takeaway: Server virtualization transforms the resources of a physical server into multiple independent virtual computing environments. It enables organizations and cloud service providers to use hardware efficiently, reduce costs, and deliver flexible, scalable computing services.

Photo of the presentation :

